Lecture 1 - Introduction

ECE 447: Data Analysis and Machine Learning for Engineers

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This Lecture Covers ...

- Introduction to AI
- Machine Learning
- Machine Learning Operations (MLOps)
- Some Insights for Machine Learning

AI Definitions - Wikipedia

Leading AI textbooks define the field as the study of "intelligent agents": any device that perceives its environment and takes actions that maximize its chance of successfully achieving its goals.

— *Wikipedia*

... Colloquially, the term "artificial intelligence" is often used to describe machines (or computers) that mimic "cognitive" functions that humans associate with the human mind, such as "learning" and "problem solving".

— *Wikipedia*

AI Definitions - Oxford Languages

The theory and development of computer systems able to perform tasks that normally require human intelligence, such as visual perception, speech recognition, decision-making, and translation between languages.

— *Oxford Languages*

AI Definitions - Bellman 1978

[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning ...

— *Bellman, 1978*

AI - Systems that ...

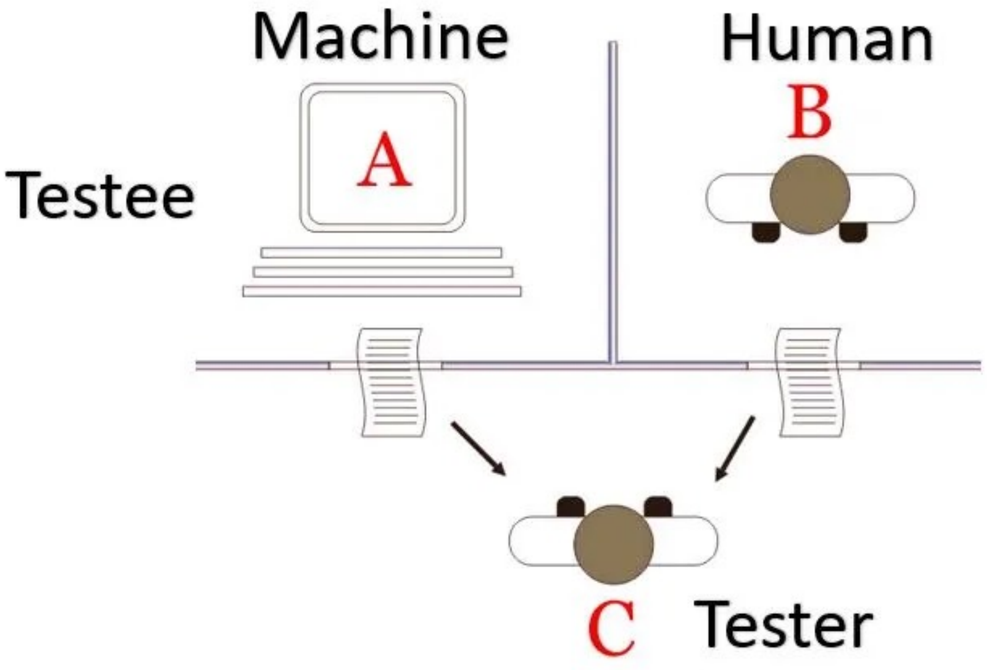
	Human-like behaviour	**Rational** behaviour
Reasoning	think humanly	think rational
Acting	act humanly	act rational

— *Artificial Intelligence - A modern approach*
Stuart J. Russell and Peter Norvig

AI - Turing Test

Alan Turing (1950) - to test if a machine acts humanly.

A large box with a man or a machine hidden in it. Written messages passed via a slot in the box. Task is to write questions and messages to the system in the box and determine if it is a human or an AI. The AI would pass the test if the human questioner cannot tell whether he is communicating with a machine or a human being. A computer that passes this test is equipped with an AI.



AI - Turing Test

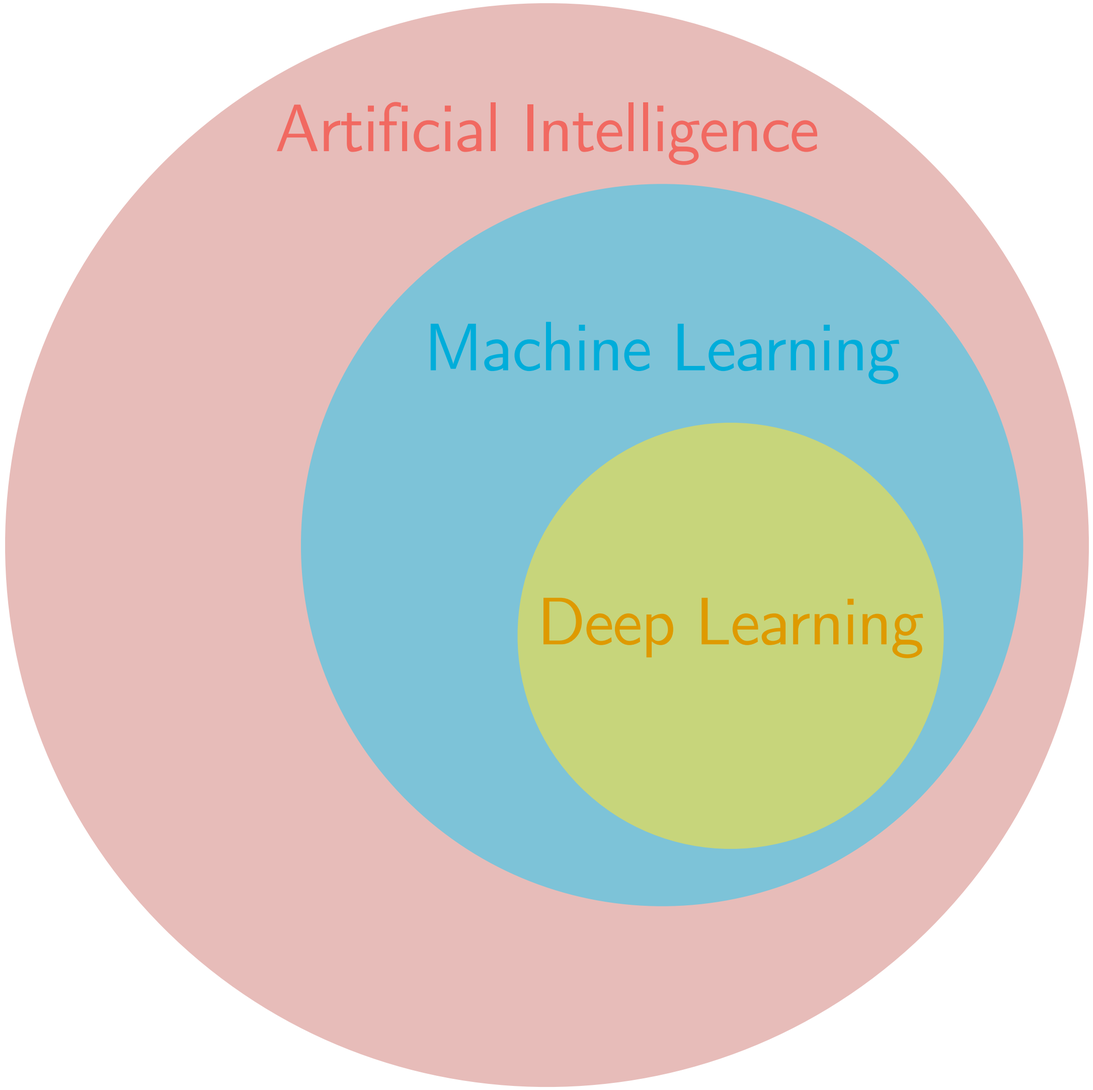
What is needed ...

- Natural Language Processing
- Knowledge Representation
- Automated Reasoning
- Machine Learning

Test in

- Computer Vision
- Robotics
- ...

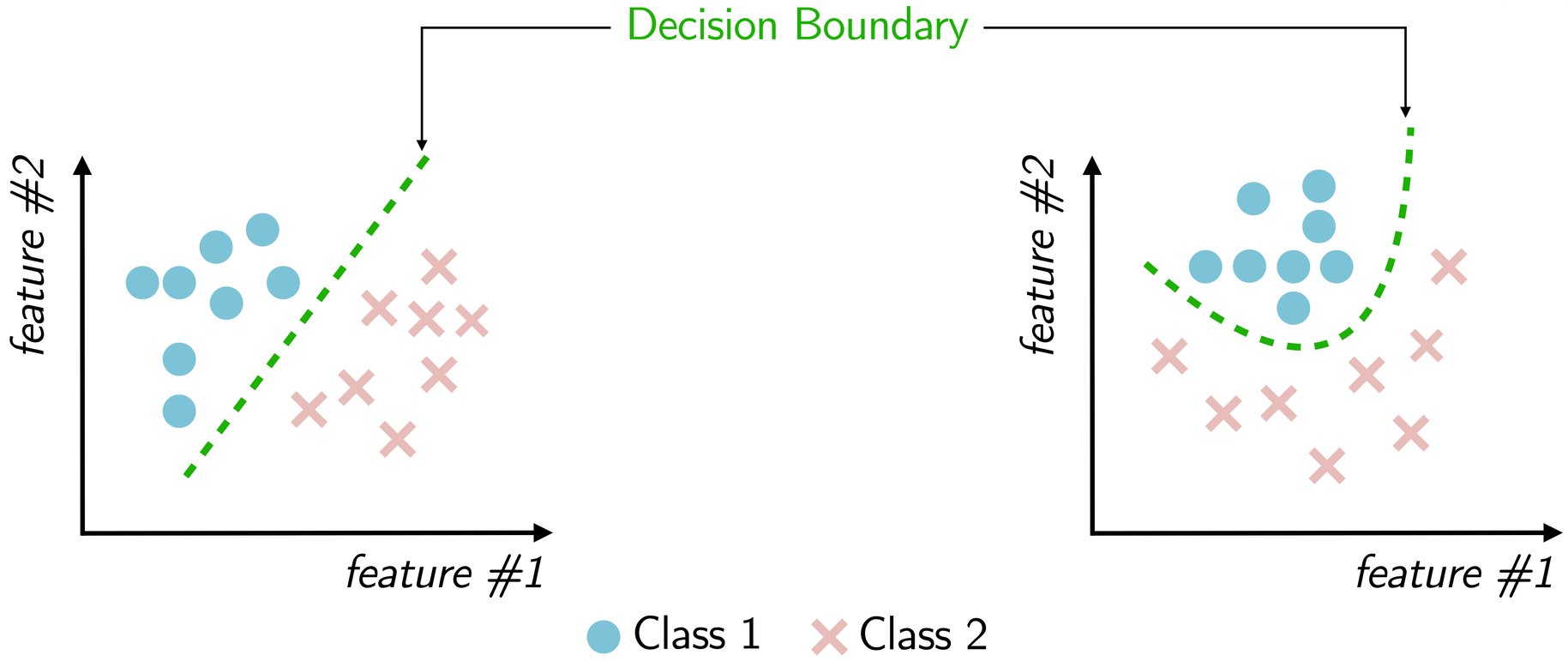
AI vs. ...



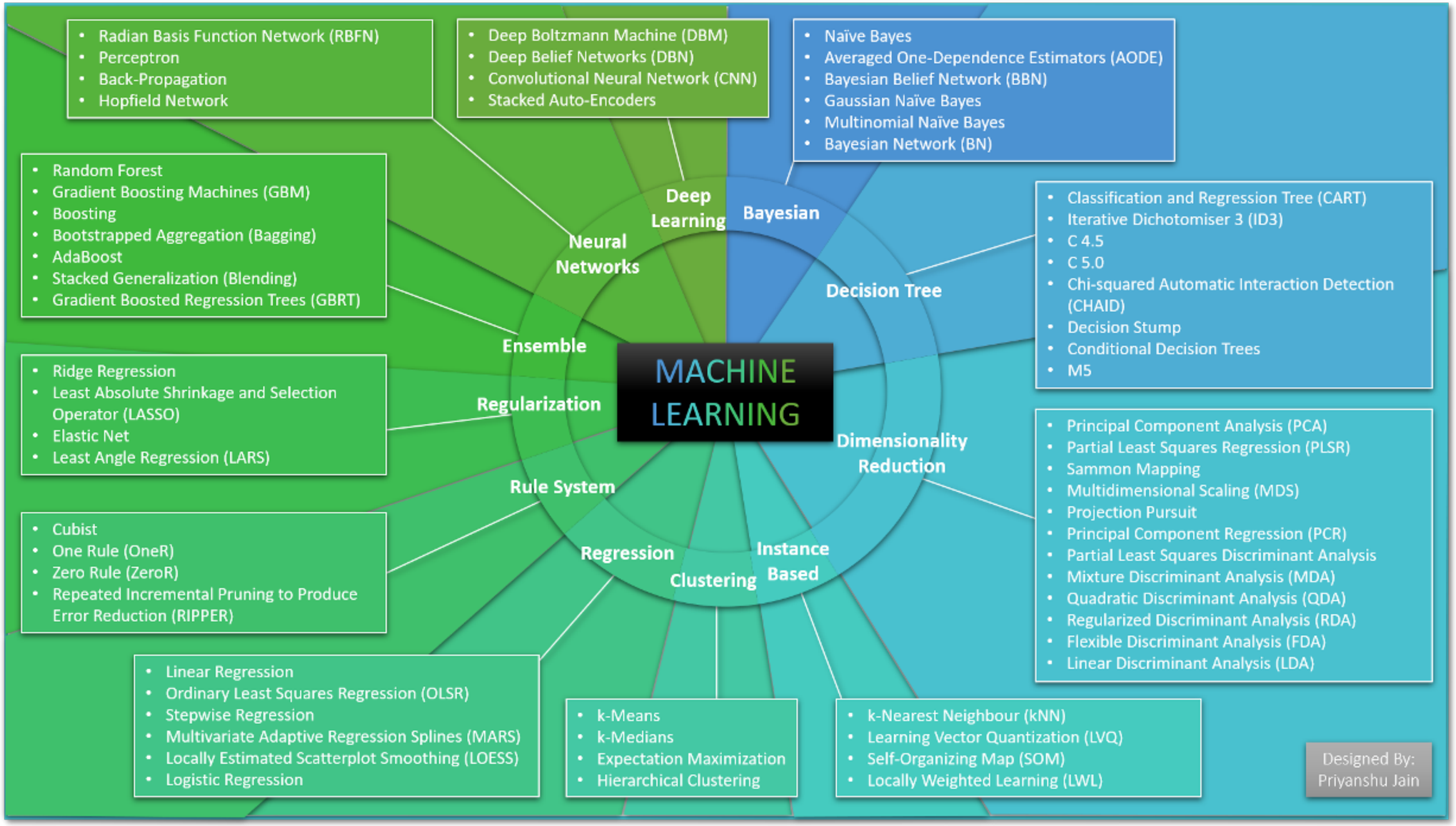
Machine Learning

The use and development of computer systems that are able to ***learn*** *and* ***adapt*** *without following* ***explicit*** *instructions, by using algorithms and statistical models to analyze and draw inferences from* ***patterns in data****.*

— *Oxford Languages*

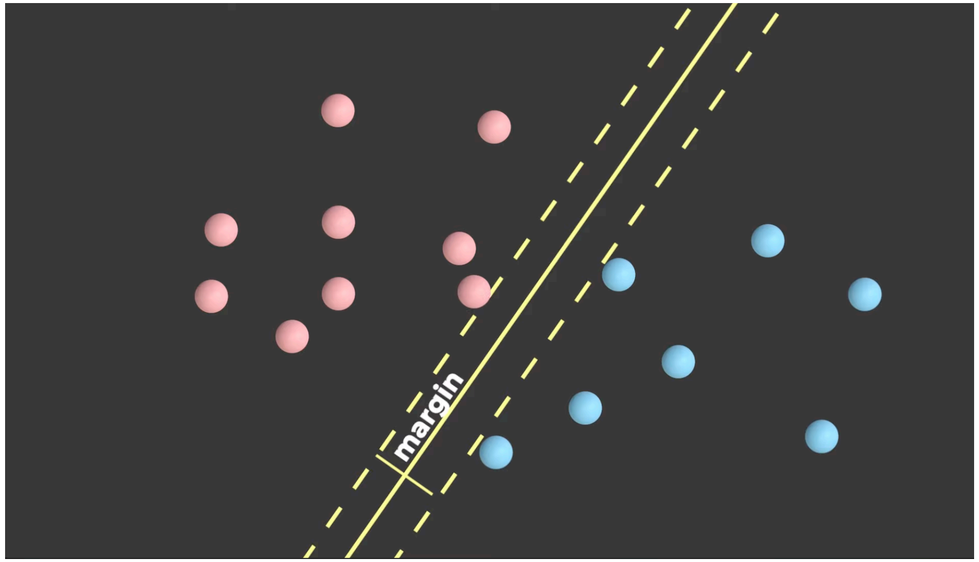


Machine Learning Methods



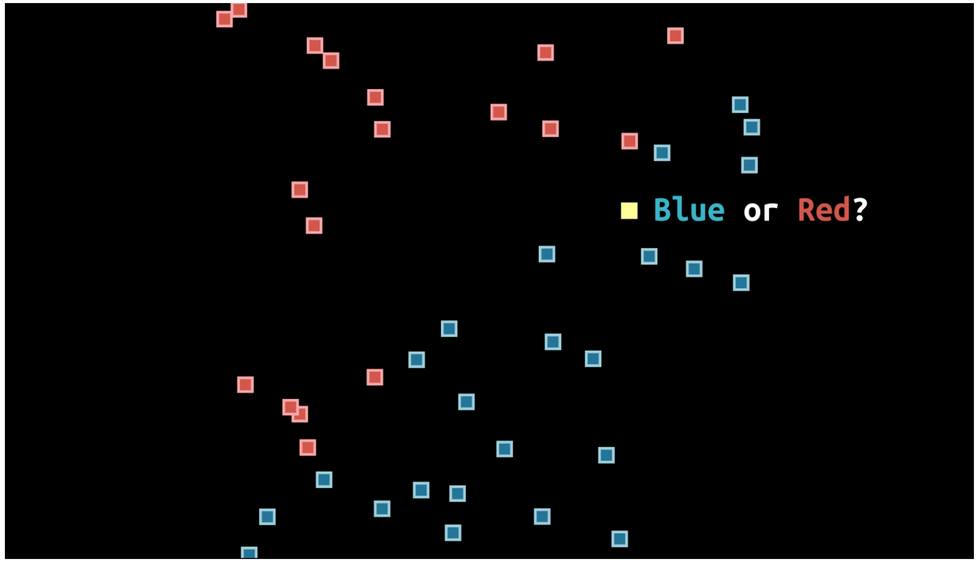
Video on SVM and KNN Machine Learning Methods

Support Vector Machine (SVM)



https://youtu.be/_YPScrckx28?si=3eZGq7oAOGqmZYKg

K-nearest Neighbours (KNN)



https://youtu.be/gs9E7E0qOIc?si=Uj1G9mimJXelle9O

Machine Learning Operations

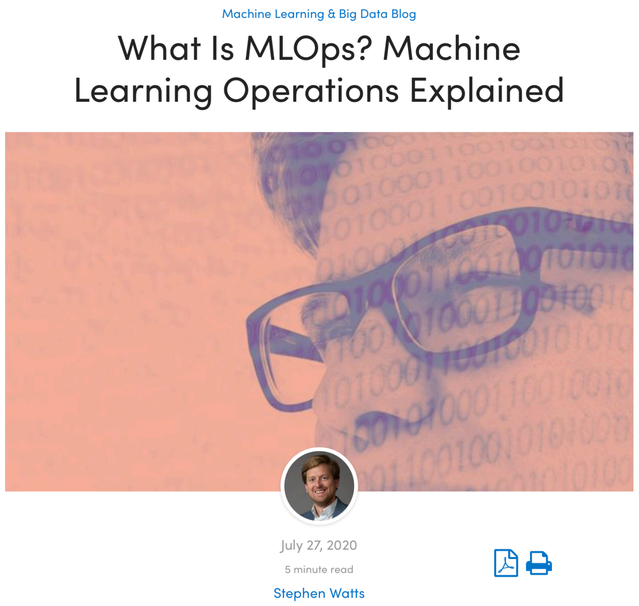
We are engineers!

A set of practices that aims to deploy and maintain machine learning models in production reliably and efficiently.

— Wikipedia

All the engineering pieces that come together and often help to deploy, run, and train AI models.

- Machine Learning
- Software Development (Dev) and IT operations (Ops)
- Data Engineering



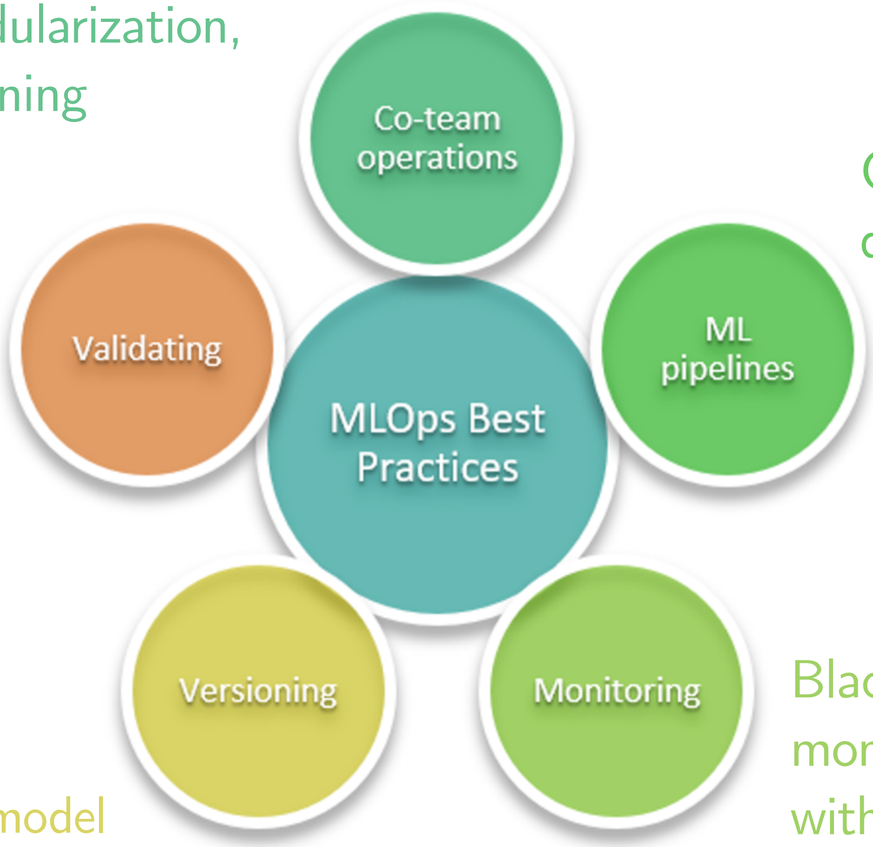
https://www.bmc.com/blogs/mlops-machine-learning-ops/

Components of Machine Learning Operations

Data scientist needs to know software Eng. practices: code modularization, reuse, testing, and versioning

Graph/flowchart representing dependencies and executions

Validation of models and data;statistical tests done on segments reflecting data



Black box requires constant monitoring to ensure staying within regulation; and programs are of quality

Versioning code + track model versions, the data used to train it, and some hyperparameters

https://www.bmc.com/blogs/mlops-machine-learning-ops/

Data Analysis Insights

- *Involves turning raw data into meaningful information;*
- *Includes collecting, processing, cleaning, and interpreting data to extract insights;*
- *Tools and techniques used in data analysis range from simple descriptive statistics to complex data mining techniques.*

Machine Learning Insights

- *Subset of AI focuses on development of algorithms able to learn from data, identify patterns, and make predictions or decisions with minimal human intervention;*

- *Automates analytical model building and allows systems to find hidden insights without being explicitly programmed where to look.*

Artificial Intelligence Insights

- *Broader concept of machines capable of performing tasks we would consider as "smart";*

- *Techniques enabling computers to mimic human intelligence, using logic, if-then rules, decision trees, and machine learning (including deep learning).*

ML as a Tool for Data Analysis

- *ML algorithms automating parts of the data analysis, such as pattern recognition and predictive modelling;*
- *ML process large volumes of data and identify trends that might not be visible to human*

AI Techniques in Data Analysis and ML

- *Enhance interpretation of data analysis results, provide context and insights;*
- *Natural Language Processing (NLP) used to automatically interpret text data and extract sentiment.*

Insights for Data Analysis and ML within AI

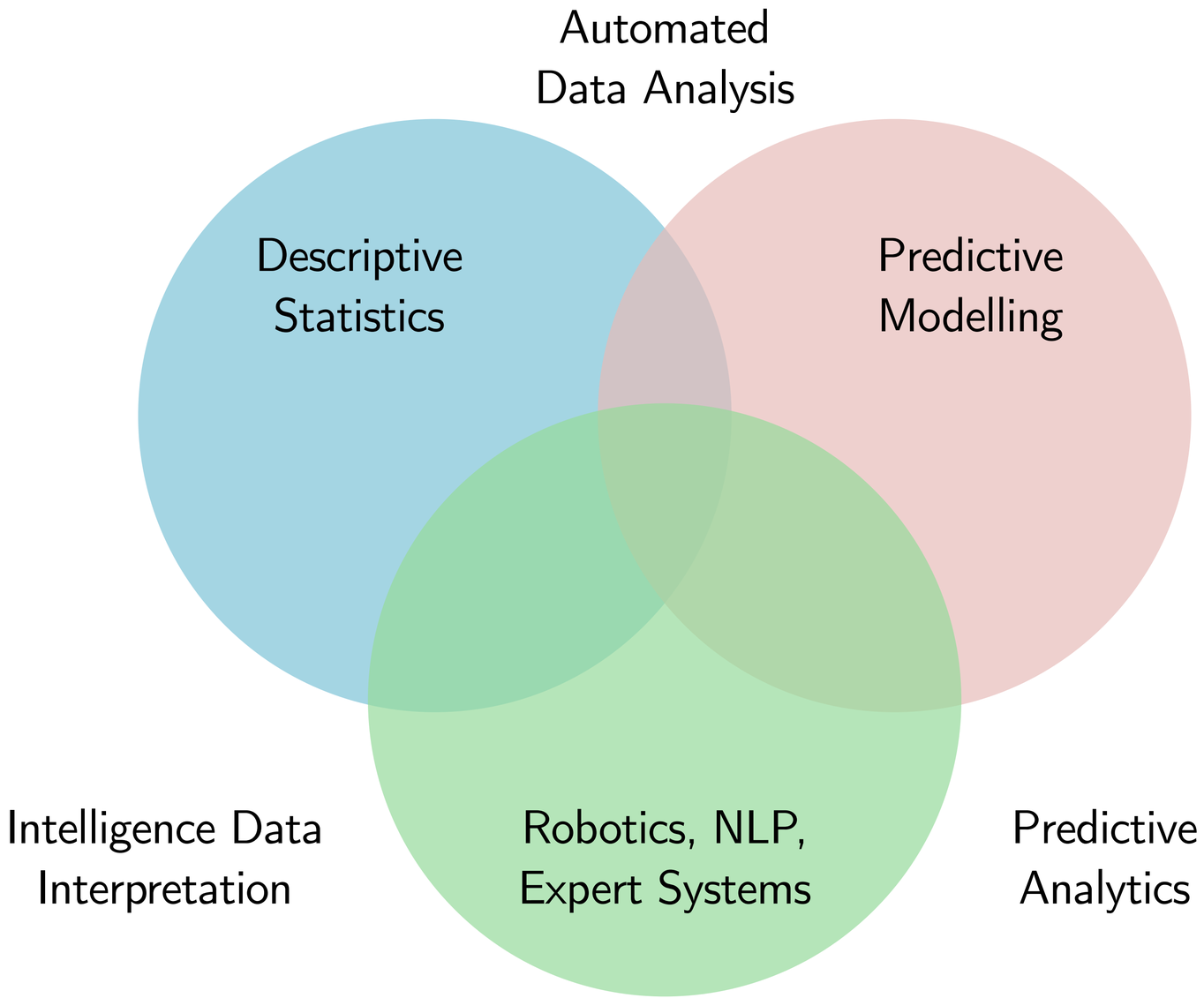
- *Predictive analytics - data analysis uses statistical algorithms and ML to identify potential future outcomes based on historical data;*

- *Data analysis techniques feed ML models, which are then used to make AI-driven predictions.*

AI-Driven Insights from Data Analysis and ML

- *AI uses outcomes/results of data analysis and ML to make or recommend decisions;*
- *AI system might analyze market trends and consumer behaviour to recommend stock purchases or sales.*

How They All Fit Together



Takeaways

- ***Artificial Intelligence*** — *Definitions are vague, imprecise.*

- ***Machine Learning*** — *Finding patterns in data and applying them.*

- ***Machine Learning Operations (MLOps)*** — *Build, test, and mange production-ready ML life cycles.*

One-Minute Paper Feedback Form (optional, not part of in-class participation)

- *What did you like the most about today's lecture? What went well?*
- *What part of today's lecture is unclear or confusing? What term or concept felt the least clear?*
- *A question you would like to ask me, but didn't get a chance during the lecture.*